Zig-Zag Sampling with Approximate Event Rates for Inferring Material Parameters

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Introduction

In computational mechanics, Bayesian inference of model parameters constitutes a non-linear inverse problem that necessitates sampling from the posterior distribution. Instead of traditional Markov Chain Monte Carlo methods, we employ a more efficient Markov process based sampler. We consider a 1d bar as a test case and want to infer its material parameters $\boldsymbol{\theta} = [E_1, E_2]^\top$. A Gaussian prior and likelihood produce a challenging non-Gaussian distribution.

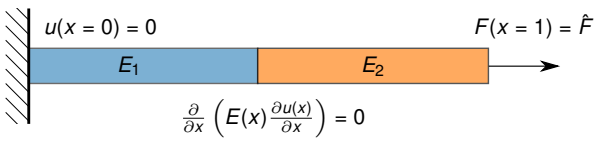

Figure 1: Schematic representation of the one-dimensional bar problem.

The Zig-Zag Process

The Zig-zag process is a Markov process with linear deterministic dynamics:

$$\boldsymbol{v} = \sum_i^d \gamma_i \boldsymbol{e}_i \quad \text{with} \quad \gamma_i = \{-1, 1\}. \tag{1}$$

One of the d velocity components is flipped at events from a Poisson process with rate

$$\lambda(\boldsymbol{\theta}, \boldsymbol{v}) = \sum_i^d \lambda_i(\boldsymbol{\theta}, \boldsymbol{v}) = \sum_i^d -\gamma_i \frac{\partial \ln \pi(\boldsymbol{\theta})}{\partial \theta_i}, \tag{2}$$

which depends on the gradient of the log-posterior density of our model parameters $\nabla \ln \pi(\boldsymbol{\theta})$. The path of the process adheres to the posterior distribution, as demonstrated in Fig. 2.

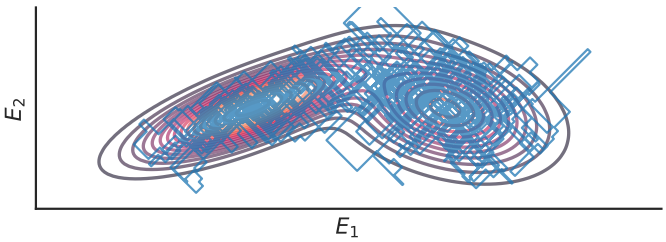

Figure 2: The Zig-Zag process with a Gaussian mixture as target.

Simulating the Events - Poisson Thinning

Instead of relying on tedious numerical integration of the exact rate, we simulate the events indirectly by thinning a homogenous Poisson process with rate $\tilde{\lambda}(\boldsymbol{\theta}, \boldsymbol{v}) \geq \lambda(\boldsymbol{\theta}, \boldsymbol{v})$. The tighter this upper bound $\tilde{\lambda}$, the more efficient the simulation.

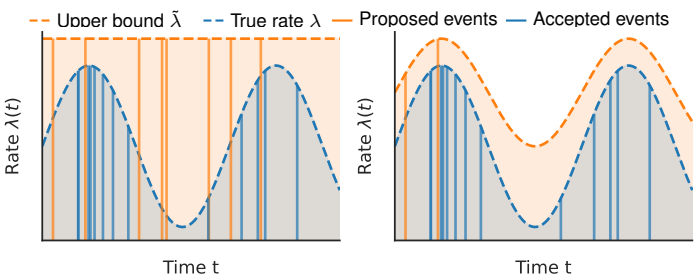

Figure 3: Poisson thinning with constant and time-dependent rates.

Thinning with Laplace Approximation

The Laplace approximation of the posterior distribution $\pi(\boldsymbol{\theta})$ provides a Gaussian approximation $\tilde{\pi}(\boldsymbol{\theta})$. The rate $\tilde{\lambda}(\boldsymbol{\theta}, \boldsymbol{v})$ is then computed as the gradient of the log-likelihood of $\tilde{\pi}(\boldsymbol{\theta})$.

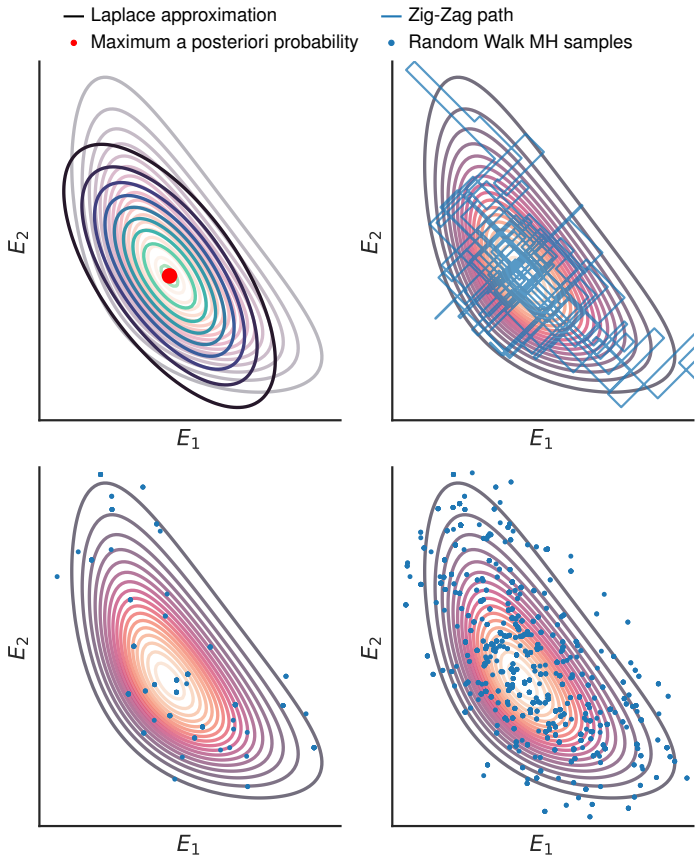

Figure 4: Zig-Zag with Poisson thinning with the Laplace approximation of the posterior and Random walk MH samples for comparison.

The simulation of the Zig-Zag process with Poisson thinning is more efficient than the numerical integration of the event rate. Further, it is significantly more accurate than the Random walk Metropolis-Hastings (MH) sampler with equal computational resources, as shown in Table 1.

Table 1: RMSE of the posterior mean for 20 runs with distinct initializations.

Sampler	Zig-Zag	Random walk MH
$N_{\text{events}} = N_{\text{samples}} = 200$	0.0022	0.0180
$N_{\text{events}} = 200 \neq N_{\text{samples}} = 2000$	0.0022	0.0173

Conclusions

This is the first application of a Markov process sampler to Bayesian inference in computational mechanics. The Zig-Zag process with Laplace thinning is more accurate than the Random walk MH sampler, demonstrating its potential. Future work will focus on finding tighter upper bounds, benchmarking in higher-dimensional settings, and the integration of error-free subsampling.

PhD progress

1	2	3	4	5